3D DESIGN AND VISUALIZATION OF A RESIDENTIAL ELECTRICAL WIRING SYSTEM

A Computer Graphics and Multimedia Project Proposal

Prepared by: Group 7

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Revised proposal

1. Introduction

Electrical wiring is an important part of residential buildings. Electrical installations are commonly represented using two-dimensional drawings and electrical symbols. However, these drawings can sometimes be difficult to visualize, especially when trying to understand how the wiring is arranged within an actual building.

This project proposes the development of a **3D model of a residential house with its electrical wiring system**. The model will show basic electrical components such as lights, switches, sockets, a distribution board, and wiring within the house.

The 3D model will provide a simple and clear way of viewing the electrical installation and will allow the wiring to be observed through different views, including cutaway views where necessary.

2. Background

Computer-Aided Design (CAD) is widely used in engineering to design and visualize physical systems before they are constructed. Three-dimensional modeling provides a better representation of the physical arrangement of components compared to conventional two-dimensional drawings.

The group plans to use FreeCAD since it is a free and open-source CAD software that can be used to create accurate three-dimensional models. It provides a suitable platform for developing a simple residential building and representing its electrical wiring system.

This project will therefore combine basic electrical installation concepts with 3D CAD modeling to produce a visual representation of a residential electrical system.

3. Literature Review

3.1 Two-Dimensional Electrical Drawings

Residential installations are traditionally documented using 2D floor-plan layouts and schematic diagrams drawn with standard graphical symbols (IEC 60617). These drawings are compact and widely accepted by electricians, but they show components as symbols on a flat plan. The reader must mentally convert the plan into three dimensions to know the height of a switch, the route of a cable inside a wall, or how circuits pass between rooms.

3.2 CAD and BIM-Based Design Tools

Professional tools such as AutoCAD Electrical are used to produce electrical schematics and panel drawings, while Building Information Modeling (BIM) tools such as Autodesk Revit (MEP) allow electrical services to be modeled in 3D together with the building [1]. These tools are powerful but are commercial, require expensive licences and powerful computers, and are aimed at professional engineers rather than learners. FreeCAD, through its Arch/BIM workbench, offers a free and open-source alternative for 3D building modeling [2].

3.3 Educational Visualization Systems

Studies on 3D virtual learning environments show that 3D representations improve spatial understanding and help learners relate abstract information to real objects [3]. Research in engineering education also shows that many students have weak spatial visualization skills, which makes interpreting 2D technical drawings difficult [4]. Virtual reality (VR) electrical training simulators exist, but they usually require special hardware and are not readily available to most students.

4. Problem Statement

Residential electrical wiring is often represented using two-dimensional drawings, which may be difficult for some users to visualize in relation to the actual structure of a building. This makes it challenging to clearly understand the physical placement of electrical components and wiring routes.

4.1 Gaps in Existing Visualization Methods

From the review above, the following gaps were identified:

- **2D drawings** do not show mounting heights or the actual cable routes inside walls and ceilings.
- **Professional CAD/BIM tools** are costly, complex and designed for engineers, not for teaching basic residential wiring.
- **VR simulators** require special hardware that most students do not have.
- There are few **free, simple 3D models** that show a complete house wiring installation based on the standards used in Uganda.

There is therefore a need for a simple three-dimensional representation that can make residential electrical wiring easier to visualize and understand.

5. Proposed Solution

The project will develop a **3D residential house model using FreeCAD** and incorporate a basic electrical wiring system into the model.

The model will include:

- Electrical sockets
- Light fittings
- Switches
- Distribution board
- Basic wiring
- Electrical conduits where appropriate

The completed model will allow users to view the house and its electrical installation from different angles. Parts of the building can also be removed or made transparent to reveal the wiring inside the walls.

6. Main Objective

To develop a 3D model for visualizing a basic residential electrical wiring system using FreeCAD.

7. Specific Objectives

1. To design a simple residential house in FreeCAD.
2. To create 3D models of basic electrical components.
3. To place the electrical components within the house model.
4. To represent the basic wiring connections in 3D.
5. To produce clear views of the completed electrical installation.

8. Scope of the Project

The project will focus on creating a **simple residential house** and representing its basic electrical installation in three dimensions.

The house may include rooms such as:

- Living room
- Bedroom
- Kitchen
- Bathroom

The electrical system will include basic components such as:

- Lights
- Switches
- Socket outlets
- Distribution board
- Basic wiring

The project will focus on **3D modeling and visualization**. It will not involve physically wiring or energizing an actual house, nor will it involve complex electrical simulations or advanced electrical calculations.

9. Wiring Standards and Practices to be Followed

The model will follow standard residential wiring practices based on **IEC 60364** and the **IET Wiring Regulations (BS 7671)**, which are commonly followed in Uganda. The main practices to be represented are:

- Single-phase 240 V, 50 Hz supply from the meter to a distribution board with a main switch, an RCD (30 mA) and MCBs.
- Separate circuits for lighting and socket outlets, each protected by its own MCB.

- Cable sizes: 1.5 mm² for lighting and 2.5 mm² for socket outlets.
- Colour coding: brown (live), blue (neutral) and green/yellow (earth).
- Typical mounting heights: switches about 1.2 m and sockets about 0.3 m above the floor.
- Cables run vertically or horizontally from accessories in PVC conduit, not diagonally across walls.
- No socket outlets inside the bathroom.

10. System Design

Figure 1 shows the overall design of the electrical system that will be modeled.

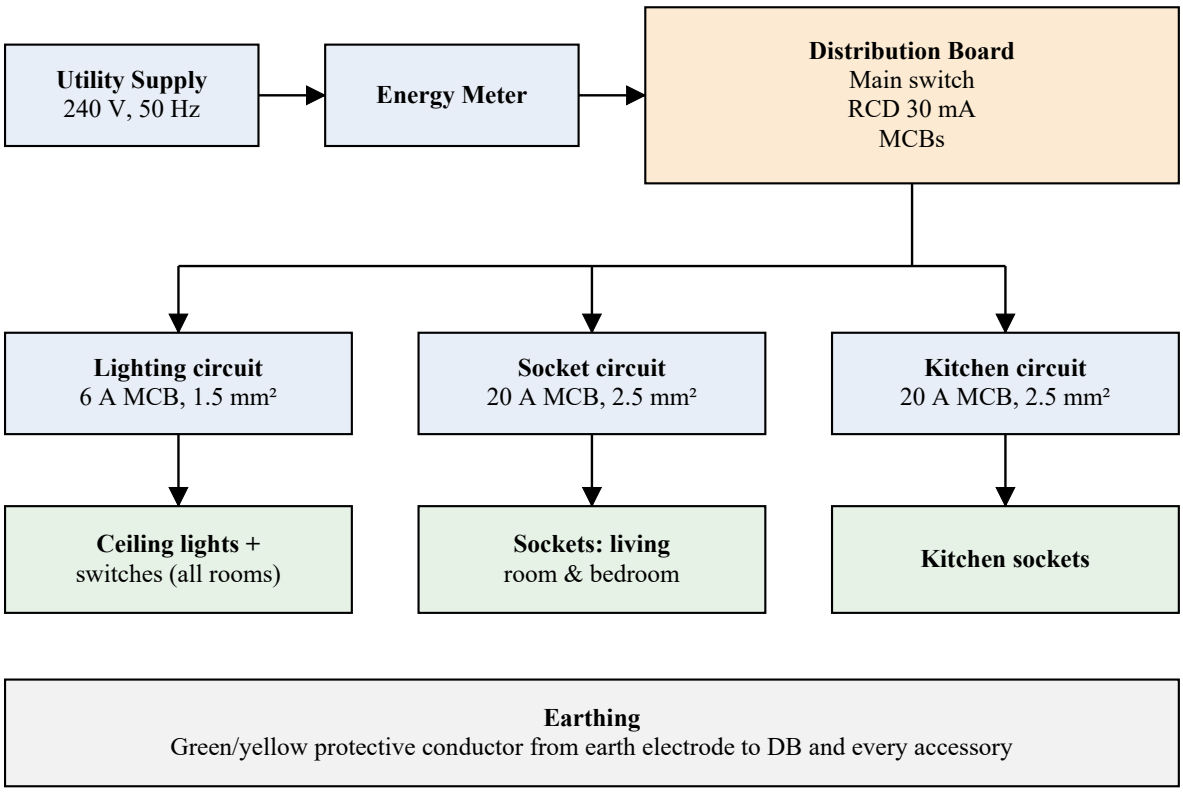


Figure 1: Block diagram of the residential electrical system to be modeled

Figure 2 shows how the 3D model will be developed, from the floor plan to the final views.

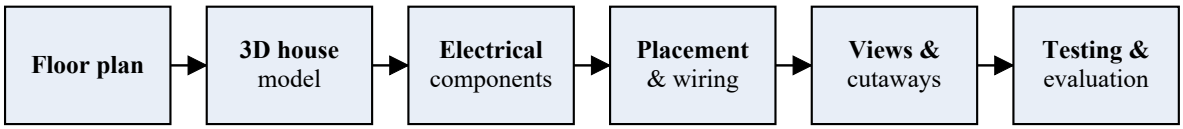


Figure 2: Model development workflow

11. Methodology

The project will be carried out through the following steps:

Step 1: House Design

A simple residential floor plan will be designed, showing the arrangement of rooms, doors and windows.

Step 2: 3D Modeling

The floor plan will be converted into a three-dimensional house model using FreeCAD.

Step 3: Electrical Component Modeling

Basic electrical components such as switches, sockets, lights and the distribution board will be created as 3D objects.

Step 4: Electrical Installation

The components will be positioned in appropriate locations within the house, following the practices in Section 9, and connected using basic 3D wiring routes.

Step 5: Visualization

Different views of the completed model will be created. A cutaway or transparent-wall view will be used to show wiring that would normally be hidden inside the walls.

Step 6: Testing and Presentation

The model will be tested and evaluated as described in Section 12. The final model will then be presented and demonstrated.

12. Testing and Evaluation Procedure

The completed model will be evaluated in three simple stages:

1. **Model check:** Each component listed in the design (Figure 1) will be checked off against the model to confirm it exists and is in the correct room. FreeCAD's *Check Geometry* tool will be used to find broken or invalid shapes.
2. **Standards check:** Mounting heights, separation of circuits and cable colours will be measured and compared with the practices in Section 9 using a checklist.
3. **User evaluation:** About 10 classmates will be shown the model and asked to trace the wiring from the distribution board to a given light or socket. They will then fill a short questionnaire rating the clarity of the model (1–5 scale) compared with a 2D wiring drawing.

Any errors found will be corrected before the final presentation.

13. Measures of Success

The project will be considered successful if the following targets are met:

Criterion	Target
All planned components modeled and placed	100% of items on the checklist
Wiring practices followed (heights, circuits, colours)	All checklist items pass

Geometry errors in the final model	None
Users who correctly trace a circuit in the model	At least 80%
Average clarity rating from users	At least 4 out of 5
Required views produced (front, top, isometric, cutaway)	All produced
Project completed on schedule	By Week 5

14. Tools and Technologies

Software

- **FreeCAD** for 3D modeling and design
- Microsoft Office word for documentation
- Canva for making presentations.

Hardware

- Laptop or desktop computers
- Computer mouse and keyboard
- Storage device for project files

15. Division of Responsibilities

The project work will be divided among the five members as follows:

Member	Responsibility
Asanansi Tracy Nankumbi	Design the house structure and floor plan
Mwambu Isaiah	Create the 3D house model in FreeCAD
Ogwapit Geresem	Create the 3D electrical components
Sarah Asingwire	Add and arrange the wiring within the house model
Walusimbi Joseph	Combine the work, create final views and prepare the presentation

The responsibilities may be changed among members during the project depending on the group's needs. All members will participate in reviewing the model, documentation, testing, and final presentation.

16. Expected Results

At the end of the project, the group expects to produce:

1. A complete 3D model of a simple residential house.
2. 3D models of basic electrical components.
3. A basic electrical wiring system integrated into the house.

4. A cutaway or transparent view showing the wiring inside the building.
5. Final 3D views of the electrical installation.
6. A project report documenting the design and development process.
7. A demonstration of the completed model.

17. Significance of the Project

The project will provide a simple and visual way of understanding residential electrical wiring. Instead of viewing the installation only through conventional two-dimensional diagrams, users will be able to see how electrical components and wiring are positioned within an actual building.

The project will also give the group practical experience in:

- 3D computer-aided design
- Electrical system visualization
- Engineering design
- FreeCAD
- Teamwork and project development

The resulting model could also be used as a basic educational demonstration tool for students learning residential electrical installation.

18. Limitations

The project will have the following limitations:

- The model will represent a basic residential electrical system.
- It will not involve physical installation or energization.
- The project will not perform advanced electrical simulations.
- The model will focus mainly on visualization rather than complete professional electrical installation design.

19. Potential Risks and Mitigation Measures

Risk	Mitigation
Limited experience with FreeCAD slows progress	Follow FreeCAD tutorials in Week 1 and start with simple shapes
Loss or corruption of project files	Keep backups on GitHub and an external storage device
Parts made by different members do not fit together	Agree on units (mm), a common origin and naming of parts before modeling
Model becomes slow due to too much detail	Keep components simple and hide parts not being worked on
Wiring layout does not follow standard practice	Use the checklist in Section 12 and consult the lecturer
Delays or a member being unavailable	Weekly progress meetings and sharing of tasks among members

20. Work Plan

Activity	Time
Project research, planning and house floor plan design	Week 1
3D house modeling and electrical component modeling	Week 2
Wiring and component placement	Week 3
Model integration, testing and evaluation	Week 4
Final views, presentation preparation, report completion and submission	Week 5

21. Conclusion

This project proposes the development of a simple **3D residential electrical wiring model using FreeCAD**. The project will combine basic residential electrical concepts with 3D computer-aided design to create a clear visualization of how electrical components and wiring are arranged within a house.

The project is designed to be achievable within the available time and resources while providing the group with practical experience in engineering design, 3D modeling, electrical visualization, and teamwork.

References

[1] C. Eastman, P. Teicholz, R. Sacks and K. Liston, *BIM Handbook: A Guide to Building Information Modeling*, 3rd ed. Hoboken, NJ: Wiley, 2018.

[2] FreeCAD Project, "FreeCAD Documentation – Arch/BIM Workbench," https://wiki.freecad.org.

[3] B. Dalgarno and M. J. W. Lee, “What are the learning affordances of 3-D virtual environments?” *British Journal of Educational Technology*, vol. 41, no. 1, pp. 10–32, 2010.

[4] S. A. Sorby, “Educational research in developing 3-D spatial skills for engineering students,” *International Journal of Science Education*, vol. 31, no. 3, pp. 459–480, 2009.

[5] IEC 60364, *Low-voltage electrical installations*. Geneva: International Electrotechnical Commission.

[6] IET, *BS 7671:2018+A2:2022 Requirements for Electrical Installations (IET Wiring Regulations)*. London: IET, 2022.